

Advanced (Carolina Reaper)**Q1.** Solve $x^2 + 4x + 4 = 0$ using the quadratic formula

- (A) $x = -2$
 (B) $x = -4$
 (C) $x = 0$
 (D) $x = 2$
 (E) $x = 4$

Q2. What is the discriminant of $x^2 - 3x - 4 = 0$?

- (A) $b^2 - 4ac = 25$
 (B) $b^2 - 4ac = 9$
 (C) $b^2 - 4ac = 1$
 (D) $b^2 - 4ac = -9$
 (E) $b^2 - 4ac = -25$

Q3. How many solutions does $x^2 + 2x + 1 = 0$ have?

- (A) 0
 (B) 1
 (C) 2
 (D) Infinitely many
 (E) None of the above

Q4. Solve $2x^2 - 5x - 3 = 0$ using the quadratic formula

- (A) $x =$
 $\frac{\sqrt{5} \pm \sqrt{49}}{4}$

- (A) $x =$
 $\frac{\sqrt{5} \pm \sqrt{25}}{4}$

- (A) $x =$
 $\frac{\sqrt{5} \pm \sqrt{14}}{4}$

- (A) $x =$
 $\frac{\sqrt{5} \pm \sqrt{9}}{4}$

(A) $x =$ $\frac{\sqrt{5} \pm \sqrt{16}}{4}$ **Q5.** What type of solutions does $x^2 + 2x + 5 = 0$ have?

- (A) Real and distinct
 (B) Real and equal
 (C) Complex conjugates
 (D) Rational
 (E) Irrational

Q6. Find the discriminant of $3x^2 - 2x - 1 = 0$

- (A) $b^2 - 4ac = 4$
 (B) $b^2 - 4ac = 16$
 (C) $b^2 - 4ac = -13$
 (D) $b^2 - 4ac = -16$
 (E) $b^2 - 4ac = 0$

Q7. Solve $x^2 - 4x - 3 = 0$ by factoring or quadratic formula

- (A) $x = -1, x = 3$
 (B) $x = 1, x = -3$
 (C) $x = 2, x = -2$
 (D) $x = -1, x = -3$
 (E) $x = 3, x = -1$

Q8. How many real solutions does $x^2 + x + 2 = 0$ have?

- (A) 0
 (B) 1
 (C) 2
 (D) Infinitely many
 (E) None of the above

Open Ended Questions (FRQ)**Q9.** Solve $x^2 + 2x - 6 = 0$ using the quadratic formula and find the discriminant. Show all work.**Q10.** Determine the number of solutions for $x^2 - 7x + 12 = 0$. Justify your answer with the discriminant and solve the equation.**Chef's Tips**

- Remind students to simplify their answers
- Encourage students to use the quadratic formula when factoring is not possible
- Emphasize the importance of checking their work